

(notes) 5.6 Rational Functions, part 1

A rational function is a polynomial divided by a polynomial

And polynomials can have a constant term, linear term, quadratic term, cubic term, etc

$$\text{rational} = \frac{\text{polynomial}}{\text{polynomial}}$$

Write some examples:

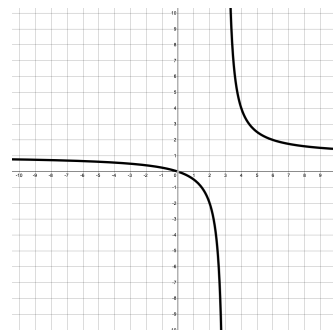
Polynomials are the perfect function because they are smooth and continuous. Their domain is all real numbers. You would think rational functions would be the same! They are smooth but they are not continuous.

Recall that whenever we have an x in the denominator, we need to put a restriction on the domain to ensure we never divide by zero!

Example. What is the domain of the function $f(x) = \frac{x}{x-3}$?

We immediately notice that there is an x in the denominator and so need to restrict the domain to prevent the denominator from ever being zero. Because as you know, when you divide by zero, you get an undefined value.

$x - 3 \neq 0 \implies x \neq 3$ and so the domain is all real numbers, except x cannot equal 3.



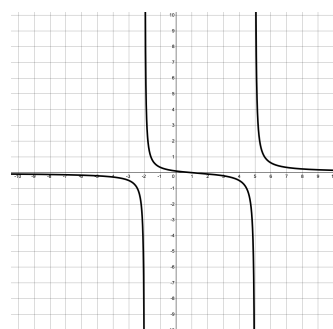
Example. What is the domain of the function $f(x) = \frac{x-1}{(x+2)(x-5)}$?

Again, the denominator cannot ever be zero! We have a factored polynomial in the denominator. What are the restrictions on the domain?

$$x + 2 \neq 0 \implies x \neq -2$$

$$x - 5 \neq 0 \implies x \neq 5$$

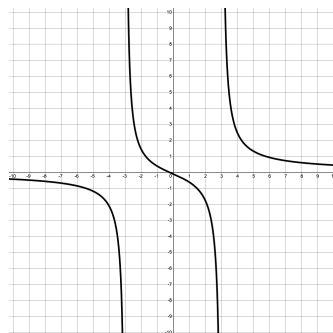
And so the domain is all real number except $x \neq -2, 5$



Example. What is the domain of the function $f(x) = \frac{4x+1}{x^2-9}$?

As if we were looking for the x intercepts of the polynomial in the denominator, we must factor first. Then find the domain restrictions.

$$\begin{aligned} x^2 - 9 &\neq 0 \\ (x - 3)(x + 3) &\neq 0 \implies x \neq 3, -3 \end{aligned}$$

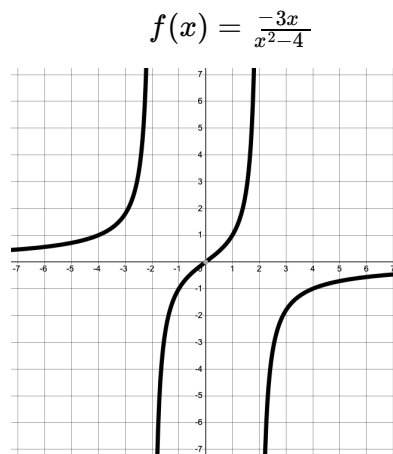
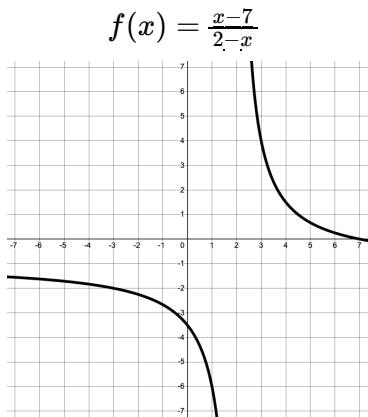
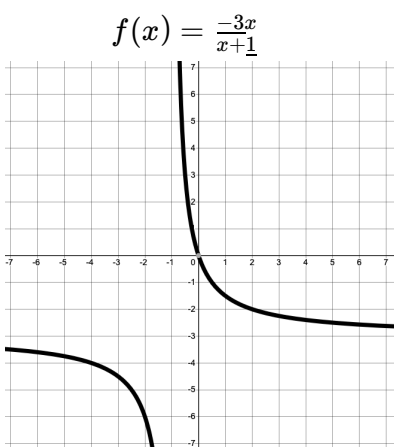


Question: Graphically, what is happening where we have domain restrictions?

Vertical Asymptote, $x =$

For the most part, rational functions will have a vertical asymptote (VA) which you will identify by finding the domain restriction (value for x that would make the denominator zero). Sometimes a function will have more than one vertical asymptote!

Practice. Draw in and write the equation for the vertical asymptotes.



Extra: Can you also identify any horizontal asymptotes?? What would be the equation for the horizontal asymptotes?

Horizontal Asymptote, $y =$

Yes, there are horizontal asymptotes (HA) most of the time. The degrees of the numerator polynomial and the denominator polynomial determine the horizontal asymptote.

For rational function $f(x) = \frac{p(x)}{q(x)}$, say the degree of numerator polynomial $p(x)$ is p and the degree of the denominator polynomial $q(x)$ is q .

Case 1: $p < q$

The numerator degree is smaller than the denominator.

Horizontal Asymptote is $y = 0$.

Ex. $f(x) = \frac{-3x}{x^2-4}$

HA: $y = 0$

Case 2: $p = q$

The numerator and the denominator has the same degree.

Horizontal Asymptote is the ratio of leading coefficients
 $y = \frac{a}{b}$.

Ex. $f(x) = \frac{2x^3-9x^2+x+1}{7x^3-1}$

HA: $y = \frac{2}{7}$

Case 3: $p > q$

The numerator has a larger degree than the denominator's degree.

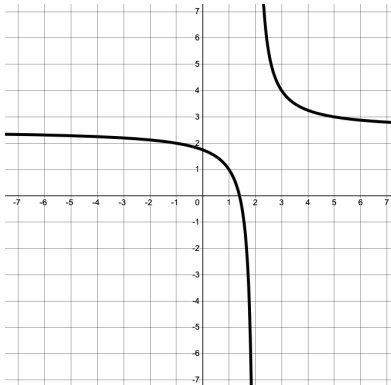
No horizontal asymptote (instead, it is a slant asymptote)

Ex. $f(x) = \frac{x^3-2}{x^2-4x+3}$

HA: none

Practice.

1. $f(x) = \frac{5x-7}{2x-4}$



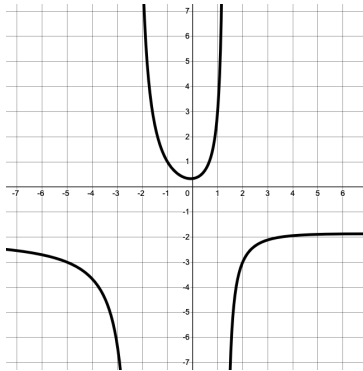
degree of top:

degree of bottom:

VA:

HA:

2. $f(x) = \frac{-2x-1}{x^2+x-3}$



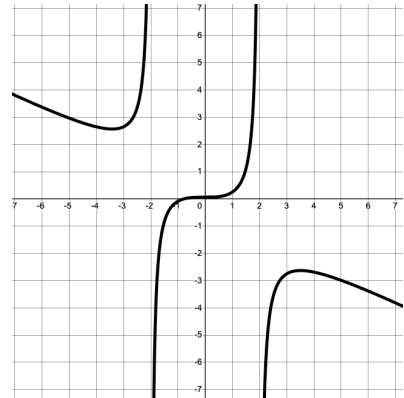
degree of top:

degree of bottom:

VA:

HA

3. $f(x) = \frac{-2x^3-1}{4x^2-16}$



degree of top:

degree of bottom:

VA:

HA:

4. $f(x) = \frac{7x^4+2x-15}{x^4-x^2}$

degree of top:

degree of bottom:

VA:

HA:

5. $f(x) = \frac{-3x^4+x-9}{x(x-4)(x+8)}$

degree of top:

degree of bottom:

VA:

HA:

6. $f(x) = \frac{x-7}{x^2+7x+12}$

degree of top:

degree of bottom:

VA:

HA: